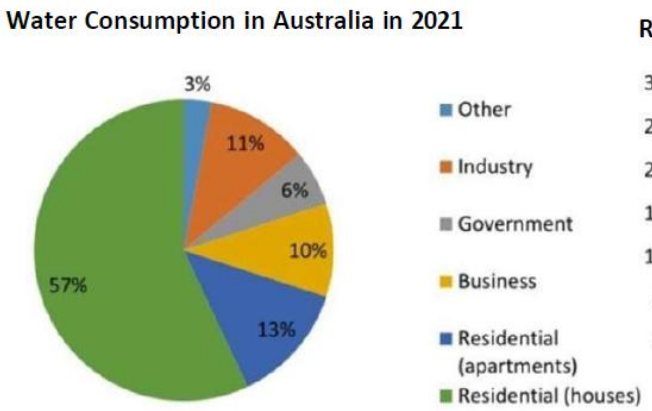
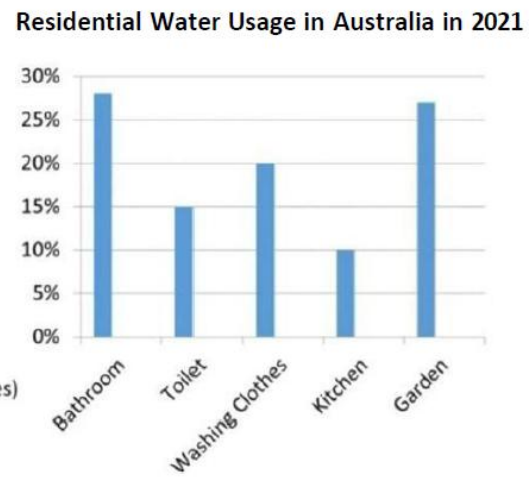
Descriptive Statistics

1. In today's data-driven world, the ability to extract meaningful insights from raw data is a fundamental skill for professionals in various fields. Your first task focuses on the analysis of two critical graphs related to water consumption in Australia. Water is a precious and finite resource and understanding its consumption patterns is vital for sustainable resource management and informed decision- making.





1.1 Analysis of Graphs

Water is one of the most important resources, and its rational use is crucial for sustainable development. Therefore, analyzing data on water consumption in Australia for 2021 helps to better understand which sectors and household habits have the greatest impact on the overall level of usage.

According to the data presented, the largest share of consumption comes from residential houses — 57% of the total volume. Apartments account for another 13%, so in total the residential sector makes up about 70%. This is a very high figure, clearly showing that it is everyday household use of water that determines the overall picture, rather than industry or business. For comparison: industry consumes 11%, business — 10%, the government sector — 6%, and the “other” category only 3%. Thus, all other sectors combined consume less water than houses and apartments alone.

If we look at the structure of consumption within the residential sector, the main “points of use” are the bathroom (28%) and the garden (27%). Together, these two categories account for more than half of all household water consumption. This can be explained both by the climatic conditions of Australia, where watering gardens and lawns is a common practice, and by cultural factors related to lifestyle. Laundry accounts for 20%, toilets for 15%, and kitchen needs for 10%.

From these data, several important conclusions can be drawn. First, it is households that create the main burden on the country’s water resources. Second, the greatest potential for savings lies in changing everyday habits and introducing simple technologies. For example, reducing shower time or using water-saving showerheads can significantly reduce bathroom consumption. Similarly, the use of rainwater collection systems and the reuse of “grey water” for irrigation can reduce the load on water supply in gardens.

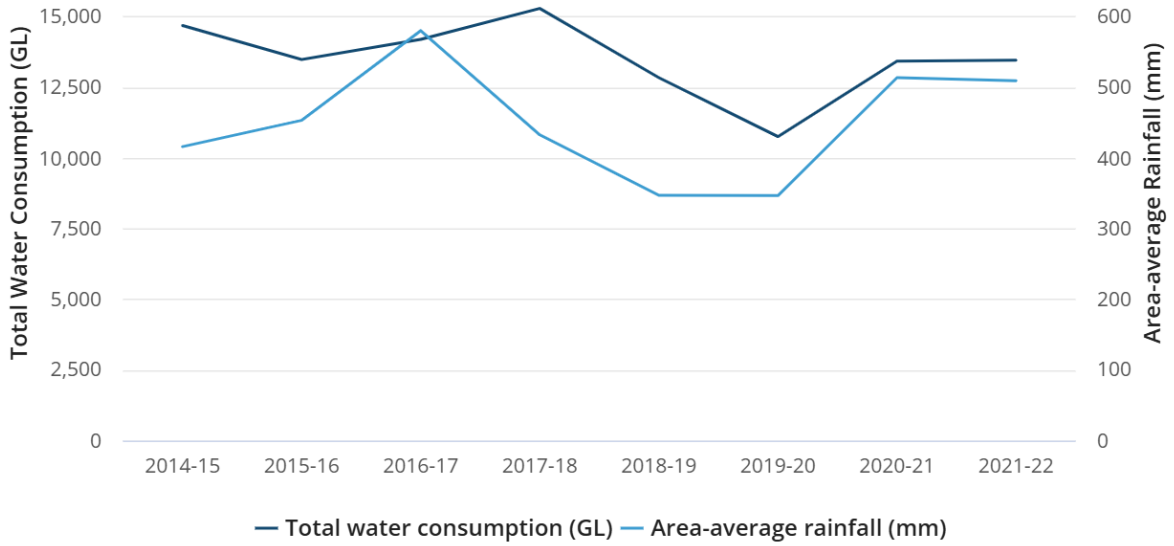
As for recommendations, they may include:

installing dual-flush toilets and water-efficient shower systems.

- using modern washing machines and dishwashers that consume less water.
- introducing rainwater harvesting systems and drip irrigation for gardens.
- conducting information campaigns to help people consciously manage water use and develop new habits.

Thus, the graphs clearly show that the main burden on Australia's water resources comes not from industry or business, but from ordinary households. Therefore, it is precisely here that efforts should be focused — both at the technological level and at the level of changing everyday behaviour.

1.2 Mean and Median



Water Account, Australia, 2021-22 financial year

https://www.abs.gov.au/statistics/environment/environmental-accounts/water-account-australia/latest-release

In statistics, measures such as the arithmetic mean and the median are often used. The mean shows the overall level of the data, while the median divides the dataset in half: half of the values are smaller, and half is larger. If the mean and the median coincide or are very close, this usually indicates that the distribution of the data is symmetric. In other words, the values are arranged roughly evenly around the centre.

However, it is important to understand that the coincidence of the mean and the median don't mean that the data are normally distributed. The normal distribution is a special case of a symmetric distribution, which has a characteristic bell-shaped form. It occurs often, but not always.

For example, the uniform distribution is also symmetric: the probability of each value is the same, and the mean equals the median. However, it does not resemble a "bell" and is not normal. There are other examples as well: a distribution may be U-shaped or have several peaks, and in such cases the mean and the median can still coincide.Therefore, the equality of these two measures can be considered only as an initial sign of symmetry, but not as proof of normality. To check whether the data really follow the normal law, additional methods are needed.The simplest way is to build a histogram. It allows a visual assessment of the shape of the distribution and helps to see whether it has a bell-shaped form. If the histogram differs greatly from the normal shape, then the coincidence of the mean and the median is not very meaningful.

Thus, the coincidence of the mean and the median is a useful signal, but for serious conclusions about normality further checks are always required.

1.3 Freshwater Withdrawals (1993–2020)

The following table contains information about the freshwater withdrawal in Australia. The data is expressed in billion cubic meters. You are required to determine the type of data that this variable is, calculate central tendency metrics and variation metrics and explain the results

```
Mean: 16.99652076321428
Median: 18.5066
Mode: 8.64
Variance: 18.83197408371831
Standard Deviation: 4.33958224760383
Range: 13.520199999999999
IQR (Interquartile Range): 7.697899999999999
```

The mean value (17.00) turned out to be lower than the median (18.51), which indicates a negative skew in the distribution: several low values in the later years pull the mean downward.
The variance and standard deviation show a moderate level of variability, meaning that the volumes of water withdrawal fluctuated noticeably from year to year.
The range (13.52) highlights a significant difference between the minimum and maximum values, while the interquartile range (7.70) shows that even the central 50% of the data cover a fairly wide spread.

Conclusion:
Thus, the data are quantitative and continuous, the distribution has a negative skew, and the variability over the observed period can be characterized as moderate.

Probabilities

2.1 It is known that in the state of Victoria, 4% of the total residential properties claim not to have access to freshwater withdrawal in their houses. In an analysis done in a subgroup last year, it is shown that, in average, 1 property did not have access to freshwater withdrawal. How many residential properties were taken to reach this result?

p= 4% of all residential properties do not have access to water.

E[X]= 1 property was found without access.

n= It is required to determine how many properties were in the subgroup.

n=1/0.04=25

There were 25 objects in the subgroup

2.2

The annual freshwater withdrawal in Sydney is normally distributed with an average of 10.5 billion cubic meters (bcm) with a standard deviation of 2 bcm. If 4 years are picked randomly, what is the probability that in two years the annual freshwater withdrawal was greater than 11 bcm?

```python
from scipy.stats import norm, binom
```

```python
# parameters of the normal distribution
mu = 10.5   # mean
sigma = 2   # standard deviation
```

```python
# probability that X > 11
p = 1 - norm.cdf(11, loc=mu, scale=sigma)
print("P(X > 11) =", round(p, 4))
```

```
P(X > 11) = 0.4013
```

```python
# binomial probability: exactly 2 successes out of 4
n = 4
k = 2
prob = binom.pmf(k, n, p)
print("P(exactly 2 out of 4 years > 11) =", round(prob, 4))
```

```
P(exactly 2 out of 4 years > 11) = 0.3463
```

The chance that in two out of four selected years the volume of water withdrawal will exceed 11 bcm is about 35%. (34.63%)